R. Jeffrey Kowalski

Data Engineer · AI Engineer · Technical Project Manager · Physicist

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Data and AI engineer with a background in physics and astrophysics and more than 20 years of experience across research labs, startups, a multinational, and Atlantic Canada's largest health research system. I build data platforms and AI tooling for medical imaging, have led product and engineering as a startup CTO, and spent my early career integrating and flying NASA balloon payloads from New Mexico and Antarctica.

EXPERIENCE

Nova Scotia Health — Biomedical Translational Imaging Centre (BIOTIC) Jul 2022 – Present
Research Analyst/Coordinator (Data & AI Engineer) Halifax, NS

- Build and orchestrate ETL pipelines in Python for academic and industry machine-learning research in diagnostic imaging (MRI, CT, ultrasound).
- Designed and implemented a Docker-based, standards-aligned DICOM de-identification toolkit (MR, CT, ultrasound) to protect personal health information, including pixel-level scrubbing of burned-in identifiers and metadata tag sanitization.
- Built ReqIntake, an autonomous requisition-management system for Nova Scotia Health: an LLM, via the Instructor library, transcribes faxed requisitions (typed and handwritten) into validated Pydantic structured output, then Python/Playwright browser automation (RPA) creates them in OceanMD and delivers them downstream to the Novari system.
- Helped build the data pipeline for a team study of a radiologist-guided, zero-shot LLM approach to MRI body protocol selection, starting from 10,207 scanned clinical requisitions; GPT-4.1 reached 87.2% top-1 and 98.3% top-3 accuracy, on par with board-certified radiologists (co-author, AIiH 2026).
- Built the data pipeline and set up the annotation system for a team study testing whether an AI segmentation model's (SAM3) performance on a clinical task can serve as an automated MRI quality metric (co-author, ISMRM 2027 submission).
- Co-led strategic planning for introducing AI into diagnostic imaging across Nova Scotia.

IWK Health Centre — BIOTIC Aug 2016 – Jul 2022
Project Manager Halifax, NS

- Managed innovation-focused imaging R&D programs, including an ~$8M ACOA Atlantic Innovation Fund program developing AI-based MRI tools in partnership with GE Healthcare Canada.

BlueLight Analytics Inc. Oct 2011 – Oct 2015
Chief Technology Officer Halifax, NS

- Led product, R&D, and engineering for a hardware-facing SaaS analytics platform for field-testing dental curing lights, serving customers in North America, South America, and Europe.

BlackBerry Dec 2008 – Jul 2011
Test Automation Developer Bedford, NS

- Developed a Python library for executing and delivering automated test cases on an international automation team.

University of Hawai'i at Mānoa — ANITA Collaboration May 2004 – Oct 2007
Graduate Research Assistant Honolulu, HI · SLAC · Fort Sumner, NM · McMurdo Station, Antarctica

- Performed hardware, firmware, communications, and RF system integration for ANITA (ANtarctic Impulsive Transient Antenna), a NASA Long Duration Balloon radio telescope.
- Participated in two NASA LDB flight campaigns, a test flight from Fort Sumner, NM and a three-month deployment to McMurdo Station on the Ross Ice Shelf, covering payload assembly, integration, and launch.

EDUCATION

M.Sc. Physics, University of Hawai'i at Mānoa 2007

B.Sc. Astrophysics, Saint Mary's University 2003

HONOURS

- **Canadian Space Agency Astronaut Recruitment Campaign** (2008–2009) — Advanced to the third round; top 100 of 5,000+ Canadian applicants
- **Antarctica Service Medal of the United States of America** (2011) — For service in Antarctica with the ANITA long-duration balloon campaign

SKILLS

Data Engineering Python, Databricks, Pandas, NumPy, SciPy, Apache Airflow, FastAPI, ETL / pipeline orchestration, Data versioning & lineage (DVC, Pachyderm), Bash

Medical Imaging & AI	DICOM & de-identification, MONAI Label, 3D Slicer, Orthanc / OHIF, Label Studio, OCR (EasyOCR), LLM structured extraction (Instructor, Pydantic), NLP (spaCy)
Cloud & DevOps	Docker / Podman, AWS (EC2, S3, RDS, IAM, Route 53), Azure, GCP, Linux administration (Ubuntu, RHEL), GitHub / GitLab